1 Time taken = 1.5 hours (or 1 hour 30 minutes) **[B1]**

$120 \div 1.5 = 80$ **[B1]**

[Total Marks: 2]

2 1 hour 12 minutes = 1.2 hours or 72 minutes **[B1]**

$\frac{102}{1.2}$ or $\frac{102}{72} \times 60$ **[M1]**

85 **[A1]**

[Total Marks: 3]

3 Time = 1.5 hours **[B1]**

Average Speed = $\frac{120}{1.5}$ **[M1]**

80 **[A1]**

[Total Marks: 3]

4 (a) M1 for $108 \div 1.5$ or $\frac{108}{90} \times 60$

72 **[A1]**

(b) M1 for $72 \times \frac{1000}{3600}$ or $72 \div 3.6$

20 **[A1]**

[Total Marks: 4]

5 (a) $\frac{120}{1.5}$ or $\frac{120}{90} \times 60$ **[M1]**

80 **[A1]**

(b) $\frac{8}{100} \times 350$ or 8×3.5 **[M1]**

28 **[A1]**

[Total Marks: 4]

6 (a) Time = 2.25 or $\frac{9}{4}$ hours **[M1]**

$180 \div 2.25 = 80$ **[A1]**

(b) Time in seconds = $450 \div 1.5 = 300$ **[M1]**

$300 \div 60 = 5$ **[A1]**

[Total Marks: 4]

7 **(a)** M1 for $\frac{30}{v} + \frac{40}{v+5} = 4$

M1 for $30(v+5) + 40v = 4v(v+5)$ (removing fractions)

A1 for obtaining $2v^2 - 25v - 75 = 0$ with no errors or omitted steps in working

(b) M2 for $(2v+5)(v-15) = 0$ or correct substitution into the quadratic formula
(M1 for $(2v+a)(v+b) = 0$ where $ab = -75$ or $2b + a = -25$)

$v = 15$ and $v = -2.5$ **[A1]**

(c) 2 **[B1]**

[Total Marks: 7]

8 Time $= 4\frac{25}{60}$ hours or 4.416... hours **[M1]**

Average Speed $= \frac{385}{4.416...}$ **[M1]**

87.2 (correct to 1 decimal place) **[A1]**

[Total Marks: 3]

9 **(a)** 45×1000 or $\frac{45}{60}$ **[M1]**

$\frac{45000}{60} = 750$ **[A1]**

(b) Base area $= 150 \times 80 = 12000$ cm^2 **[M1]**

Rate of height increase $= \frac{\text{Volume per second}}{\text{Base area}} = \frac{750}{12000}$ **[M1]**

0.0625 **[A1]**

[Total Marks: 5]

10 **(a)** $\frac{350}{100} \times 6.4$ **[M1]**

22.4 **[A1]**

(b) Cost in euros $= 22.4 \times 1.35 = 30.24$ **[M1]**

Cost in dollars $= 30.24 \div 0.85$ **[M1]**

35.58 **[A1]**

[Total Marks: 5]

11 **(a)** $150 \times 60 \times 50$ or 450000 seen or $1.5 \times 0.6 \times 0.5 = 0.45$ m^3 **[M1]**

450 **[A1]**

(b) $\frac{450}{24}$ or 18.75 **[M1]**

0.75×60 or 45 **[M1]**

18 minutes 45 seconds **[A1]**

[Total Marks: 5]

12 **(a)** M1 for 52×1000 or 52000 seen

M1 for their $52000 \div 1.6$ (implies 32500 seconds)

9 hours 2 minutes **[A1]**

(b) M1 for 52×2.15 or 111.8 seen

M1 for their 111.8×0.84

93.91 **[A1]**

[Total Marks: 6]

13 **(a)** Fuel used $= \frac{325}{100} \times 8.4$ or 27.3 **[M1]**

27.3×1.35 **[M1]**

36.855 or 36.86 **[A1]**

(b) 24 minutes = 0.4 hours or $\frac{2}{5}$ hours **[B1]**

Distance $= 75 \times 4.4$ **[M1]**

330 **[A1]**

[Total Marks: 6]

14 **(a)** Volume in $m^3 = 1.8 \times 1.5 \times 1.2 = 3.24$ **[M1]**

$3.24 \times 1000 = 3240$ litres **[A1]**

(b) Time in seconds $= \frac{3240}{2.5} = 1296$ **[M1]**

Time in minutes $= \frac{1296}{60} = 21.6$ **[A1]**

(c) Rate $= \frac{\text{Volume}}{\text{Time}} = \frac{3240}{45}$ **[M1]**

$x = 72$ **[A1]**

[Total Marks: 6]

15 **(a)** Volume $= 2.5 \times 1.4 \times 1.2$ or 4.2 (m^3) **[M1]**

Volume in litres $= 4.2 \times 1000$ or 4200 **[M1]**

Time in seconds $= \frac{4200}{0.35}$ or 12000 **[M1]**

3 h 20 min **[A1]**

(b) $\frac{780}{0.84}$ or 928.57... **[M1]**

$928.57 - 850$ **[M1]**

78.57 **[A1]**

[Total Marks: 7]

16 **(a)** Time A to B $= \dfrac{d}{v}$ **[M1]**

Time B to C $= \dfrac{2d}{v+15}$ **[M1]**

Total Time $T = \dfrac{d}{v} + \dfrac{2d}{v+15} = \dfrac{d(v+15)+2dv}{v(v+15)} = \dfrac{dv+15d+2dv}{v(v+15)} = \dfrac{3dv+15d}{v(v+15)}$ **[A1]**

(b) Substitute $d = 120$ and $T = 4.5$ into the equation from part (a): $4.5 = \dfrac{3(120)v + 15(120)}{v(v+15)}$ **[M1]**

$4.5(v^2 + 15v) = 360v + 1800$ **[M1]**

$4.5v^2 + 67.5v - 360v - 1800 = 0$ **[M1]**

$4.5v^2 - 292.5v - 1800 = 0 \implies divideby1.5toget3\mathrm{v}^2 - 195v - 1200 = 0$ **[A1]**

[Total Marks: 7]

17 **(a)** $\dfrac{x}{12} + \dfrac{x+5}{15} = \dfrac{31}{12}$ or $2\dfrac{35}{60}$ [M2]
(M1 for either $\dfrac{x}{12}$ or $\dfrac{x+5}{15}$ seen as a time)

$\dfrac{5x + 4(x+5)}{60} = \dfrac{155}{60}$ or equivalent correct clearing of fractions **[M1]**

$x = 15$ **[A1]**

(b) Total distance $= 15 + (15 + 5) = 35$ **[B1]**

Average speed $= \dfrac{35}{\frac{31}{12}}$ **[M1]**

13.5 **[A1]**

[Total Marks: 7]

18 **(a)** $\dfrac{2400}{v}$ **[B1]**

(b) $\dfrac{2400}{v} + \dfrac{2400}{v+40} = 10.5$ **[M1]**

$2400(v + 40) + 2400v = 10.5v(v + 40)$ **[M1]**

$2400v + 96000 + 2400v = 10.5v^2 + 420v$ **[M1]**

$10.5v^2 - 4380v - 96000 = 0$ **[A1]**

Multiply by $\dfrac{2}{3}$ to get $7v^2 - 2920v - 192000 = 0$ **[A1]**

(c) $v = \dfrac{-(-2920) \pm \sqrt{(-2920)^2 - 4(7)(-192000)}}{2(7)}$ **[M1]**

$v = 475.25...$ or $v = -57.68...$ (discard negative) **[A1]**

475 (correct to nearest integer) **[A1]**

[Total Marks: 7]

19 **(a)** $\dfrac{30}{x} + \dfrac{18}{x-4} = 2.5$ **[M1]**

$\dfrac{30(x-4)+18x}{x(x-4)} = 2.5$ **[M1]**

$48x - 120 = 2.5(x^2 - 4x)$ **[M1]**

$48x - 120 = 2.5x^2 - 10x \implies 2.5x^2 - 58x + 120 = 0 \implies 5x^2 - 116x + 240 = 0$ **[A1]**

(b) $x = \dfrac{-(-116) \pm \sqrt{(-116)^2 - 4(5)(240)}}{2(5)}$ **[M1]**

$x = \dfrac{116 \pm \sqrt{13456 - 4800}}{10} = \dfrac{116 \pm \sqrt{8656}}{10}$ **[M1]**

$x = 20.90$ or $x = 2.30$ **[A1]**

[Total Marks: 7]

20 **(a)** $\dfrac{15}{x} + \dfrac{12}{x-4} = 3.5$ **[M1]**

Multiply by $2x(x-4)$ to clear denominators (or equivalent): $30(x-4) + 24x = 7x(x-4)$ **[M1]**

$54x - 120 = 7x^2 - 28x \implies 7x^2 - 82x + 120 = 0$ **[A1]**

(b) **(i)** $(7x - 12)(x - 10) = 0$ OR $\dfrac{82 \pm \sqrt{(-82)^2 - 4(7)(120)}}{14}$ **[M2]**

If M0, allow M1 for $(7x \pm a)(x \pm b) = 0$ where $ab = 120$ or $7b \pm a = -82$

$x = 10$, $x = \dfrac{12}{7}$ (or 1.71) **[A1]**

(ii) Identifies valid speed $x = 10$ (since $x - 4 > 0 \implies x > 4$) **[B1]**

Time $= \dfrac{15}{10} = 1.5$ **[B1]**

[Total Marks: 8]

21 **(a)** $\dfrac{150}{x}$ **[B1]**

(b) $\dfrac{150}{x} + \dfrac{90}{x+15} = 3$ **[M1]**

$\dfrac{150(x+15)+90x}{x(x+15)} = 3$ or $150(x+15) + 90x = 3x(x+15)$ **[M1]**

$150x + 2250 + 90x = 3x^2 + 45x$ **[M1]**

$3x^2 - 195x - 2250 = 0$ leading to $x^2 - 65x - 750 = 0$ **[A1]**

(c) $(x-75)(x+10)=0$ or $\dfrac{65 \pm \sqrt{(-65)^2-4(1)(-750)}}{2}$ **[M2]**

If M0 scored, allow [M1] for $(x \pm 75)(x \pm 10)=0$ or for correct substitution into the quadratic formula.

75 **[A1]**

[Total Marks: 8]